

WINSTON-SALEM REYNOLDS, NC, USA

WMO: 723193

Lat: 36.134N

Lon: 80.222W

Elev: 970

StdP: 14.19

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93807

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 19.1	(c) 23.1	(d) -1.8	(e) 5.2	(f) 25.7	(g) 2.6	(h) 6.5	(i) 29.6	(j) 20.5	(k) 37.7	(l) 18.6	(m) 39.1	(n) 6.0	(o) 320	(p) 0.389

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	92.2	73.5	90.3	73.0	88.3	72.4	76.4	86.7	75.3	85.1	74.5	83.7	6.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.3	128.3	79.9	72.5	124.7	79.2	71.9	122.3	78.7	40.4	86.9	39.3	85.7	38.5	83.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 17.5	(b) 15.2	(c) 12.7	(e) 12.4	(f) 95.5	(g) 5.1	(h) 2.6	(i) 8.8	(j) 97.4	(k) 5.8	(l) 98.9	(m) 3.0	(n) 100.4	(o) -0.7	(p) 102.2		
			DB			2.6	7.7	80.6	5.3	82.3	3.1	84.0	0.1	86.1		
			WB			4.1	2.9									

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.9	39.7	43.3	50.6	60.1	68.0	75.5	78.2	77.2	70.8	60.2	50.4	43.6
	DBStd	15.48	10.26	9.21	9.52	8.23	6.76	4.86	4.17	4.32	5.94	7.70	8.41	8.57
	HDD50	1044	346	226	113	14	0	0	0	0	0	11	97	237
	HDD65	3398	783	607	455	189	52	2	0	0	19	186	442	663
	CDD50	4655	28	39	130	317	558	765	875	842	625	328	108	40
	CDD65	1535	0	1	7	42	144	317	410	377	194	39	3	1
	CDH74	13051	1	3	57	398	1140	2788	3773	3254	1339	277	20	1
	CDH80	4650	0	0	3	72	303	1052	1539	1226	402	52	1	0
Wind	WSAvg	5.4	5.9	6.1	6.8	6.6	5.6	5.0	4.6	4.3	4.9	4.9	5.2	5.2
Precipitation	PrecAvg	45.9	3.7	2.8	4.3	3.7	3.7	4.4	4.8	4.3	4.4	3.3	3.0	3.5
	PrecMax	63.6	6.8	5.3	8.4	7.0	8.3	8.1	8.1	7.9	9.4	12.5	7.3	6.5
	PrecMin	32.7	0.8	1.2	0.8	0.5	1.1	0.9	2.5	1.4	0.6	0.0	0.4	1.2
	PrecStd	6.4	1.6	1.1	2.0	1.8	1.7	1.7	1.5	1.5	2.3	2.8	1.7	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.9	73.0	79.5	85.7	89.6	93.9	95.2	94.1	91.0	85.3	77.1	72.0
		MCWB	59.7	60.9	59.5	64.2	69.9	72.2	74.6	74.0	69.7	70.1	62.3	60.6
	2%	DB	64.3	67.5	74.9	82.0	86.2	91.0	92.2	91.2	87.7	80.5	72.3	66.2
		MCWB	56.5	56.3	58.9	63.3	68.6	72.7	74.2	73.9	70.8	66.7	59.8	59.8
	5%	DB	60.8	63.6	71.0	79.0	83.5	88.7	90.2	89.3	84.6	76.9	68.7	62.3
		MCWB	53.6	54.0	56.4	61.9	68.1	72.0	73.6	73.4	69.7	64.2	59.3	55.8
	10%	DB	55.9	59.1	66.3	75.2	80.8	86.1	88.0	86.5	81.6	73.2	64.8	57.8
		MCWB	48.7	49.1	54.2	60.5	66.6	70.7	73.1	72.6	68.9	63.4	56.7	51.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.0	63.7	64.7	69.0	73.3	76.6	77.8	78.2	75.4	73.4	67.4	64.9
		MCDB	66.7	69.3	74.0	79.0	83.1	86.9	89.6	88.0	83.5	78.2	70.2	68.6
	2%	WB	59.1	59.8	61.9	66.8	71.3	75.0	76.6	76.3	73.9	70.9	64.0	61.0
		MCDB	63.1	65.5	69.9	77.1	81.1	85.9	87.6	86.0	81.3	77.2	68.4	64.7
	5%	WB	54.7	55.8	59.6	64.5	69.8	73.7	75.4	75.2	72.6	68.0	61.6	57.2
		MCDB	59.3	61.4	67.6	73.9	79.1	84.3	85.7	84.3	79.2	73.8	67.0	61.1
	10%	WB	49.9	51.1	56.9	62.6	68.5	72.7	74.6	74.3	71.3	65.2	58.2	52.6
		MCDB	55.6	57.1	64.3	71.7	77.3	82.5	84.3	82.9	77.3	71.5	63.8	56.4
Mean Daily Temperature Range	5% DB	MDBR	17.3	18.5	19.4	20.7	19.2	18.0	17.0	16.7	17.1	19.2	19.7	16.7
		MCDBR	21.5	23.8	23.4	24.1	22.0	20.8	19.5	19.6	20.6	22.7	23.5	20.4
		MCWBR	14.9	16.0	12.6	10.6	8.5	7.3	6.4	6.4	7.6	10.9	14.0	14.4
	5% WB	MCDBR	19.0	20.8	20.7	20.5	18.1	18.3	17.5	16.6	16.3	17.2	18.5	17.4
		MCWBR	15.7	16.9	12.9	10.5	8.1	7.0	6.6	6.3	7.2	10.2	14.4	14.4
Clear-Sky Solar Irradiance	taub	0.304	0.318	0.348	0.391	0.433	0.470	0.512	0.504	0.409	0.365	0.320	0.306	
	taud	2.506	2.462	2.420	2.304	2.213	2.147	2.057	2.057	2.321	2.423	2.508	2.534	
	Ebn at Noon	285	293	292	283	271	260	248	247	268	272	276	277	
	Edh at Noon	26	30	35	41	45	49	53	52	38	31	25	24	
All-Sky Solar Radiation	RadAvg	800	1009	1345	1681	1858	1998	1861	1703	1426	1167	885	683	
	RadStd	53	123	89	134	153	146	122	86	131	146	72	61	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.70	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (58 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+250	+166

Nomenclature: See separate page